

Yiyang Su

✉ suyiyan1@msu.edu | 🌐 <https://yiyang-su.github.io> | 💻 [yiyang-su](#) | 🎓 Yiyang Su

Education

Michigan State University

PH.D. IN COMPUTER SCIENCE

- Advised by Dr. Xiaoming Liu
- Anticipated graduation date: Fall 2026

East Lansing, MI
2021 - present

University of Rochester

B.S. IN COMPUTER SCIENCE AND B.S. IN MATHEMATICS

- Highest Distinction in Computer Science, High Honors in Research in Computer Science
- Highest Distinction in Mathematics, Honors in Research in Mathematics
- GPA: 3.96/4.0; Magna Cum Laude; Dean's List for all eligible semesters

Rochester, NY
2017 - 2021

Honors & Awards

- | | | |
|------|---|---------------------------|
| 2021 | Engineering Distinguished Scholar , 1 year of support awarded to outstanding incoming PhD students | Michigan State University |
| 2021 | Phi Beta Kappa , elected member through the Iota of New York Chapter | University of Rochester |
| 2021 | Donald M. Barnard Prize , awarded to 10 junior/senior engineering students based on merit | University of Rochester |

Publications

- [1] **Yiyang Su**, Jie Zhu, Feng Liu, Anil Jain, and Xiaoming Liu. "SapiensID 2.0: Aligning Human Recognition Foundation Models with Human Perception". *Advances in Neural Information Processing Systems (NeurIPS)*. 2026.
- [2] Xiang Xu, **Yiyang Su**, Eric Zhao, Zheng Zhang, Zhuowen Tu, Anil Jain, and Jonathan Wu. "Beyond Disjoint Tasks: Towards More Natural Continual Learning for Vision-Language Models". *Proceedings of the European Conference on Computer Vision (ECCV)*. 2026.
- [3] Jie Zhu, Xiao Guo, **Yiyang Su**, Anil Jain, and Xiaoming Liu. "FusionAgent: A Multimodal Agent with Dynamic Model Selection for Human Recognition". *Proceeding of IEEE Computer Vision and Pattern Recognition (CVPR)*. 2026.
- [4] Jie Zhu, **Yiyang Su**, and Xiaoming Liu. "Can Textual Reasoning Improve the Performance of MLLMs on Fine-grained Visual Classification?" *Proceeding of Findings of IEEE Computer Vision and Pattern Recognition (CVPR)*. 2026.
- [5] **Yiyang Su**, Feng Liu, and Xiaoming Liu. "Hierarchical and Adaptive Mixture of Biometric Experts for Video-based Person ReID". *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)*. 2025.
- [6] Jie Zhu, **Yiyang Su**, Minchul Kim, Anil K. Jain, and Xiaoming Liu. "A Quality-Guided Mixture of Score-Fusion Experts Framework for Whole-Body Biometric Recognition". *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)*. 2025.
- [7] Minchul Kim, Dingqiang Ye, **Yiyang Su**, Feng Liu, and Xiaoming Liu. "SapiensID: Foundation for Human Recognition". *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2025.
- [8] **Yiyang Su**, Minchul Kim, Feng Liu, Anil K. Jain, and Xiaoming Liu. "Open-Set Biometrics: Beyond Good Closed-Set Models". *Proceedings of the European Conference on Computer Vision (ECCV)*. 2024.
- [9] Minchul Kim, Feng Liu, **Yiyang Su**, Anil K. Jain, and Xiaoming Liu. "KeyPoint Relative Position Encoding for Face Recognition". *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2024.
- [10] Feng Liu, Ryan Ashbaugh, Nicholas Chimitt, Najmul Hassan, Ali Hassani, Ajay Jaiswal, Minchul Kim, Zhiyuan Mao, Christopher Perry, Zhiyuan Ren, **Yiyang Su**, Pegah Varghaei, Kai Wang, Xingguang Zhang, Stanley Chan, Arun Ross, Humphrey Shi, Zhangyang Wang, Anil Jain, and Xiaoming Liu. "FarSight: A Physics-Driven Whole-Body Biometric System at Large Distance and Altitude". *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 2024.
- [11] Zhiyuan Ren, **Yiyang Su**, and Xiaoming Liu. "ChatGPT-Powered Hierarchical Comparisons for Image Classification". *Advances in Neural Information Processing Systems (NeurIPS)*. 2023.
- [12] **Yiyang Su**, Ali Vosoughi, Shijian Deng, Yapeng Tian, and Chenliang Xu. "Separating Invisible Sounds toward Audio-Visual Scene-Aware Separation". *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*. 2023.

Professional Experience

Applied Scientist Intern

AMAZON AGI

- Credit assignment for self-evolving LLM agents in long-horizon settings.

Bellevue, WA
May 2026 - Aug. 2026

Applied Scientist Intern

AWS AI LABS

- In light of the observation that existing continual learning benchmarks for vision-language models are based on disjoint tasks, while in reality, different tasks may exhibit overlap and interference, proposed a novel benchmark to address this gap. Demonstrated that the existing model encounters challenges in this more realistic scenario and propose a new framework incorporating dedicated LoRA modules for each task and an agentic router to route each sample to the most suitable LoRA module [2].

Seattle, WA
May 2025 - Sep. 2025

Research Assistant

COMPUTER VISION LAB

- Developed a novel loss function aligned with the objectives of open-set biometrics to address the misalignment of existing loss functions. Applied this loss function to face/gait recognition and person reID and achieved an improvement of up to 2.9% on closed-set and 4.2% on open-set performance [8].
- Created a video-based person reID framework that independently models appearance, body shape, and pose features and adaptively integrate them using a hierarchical mixture of expert model based on the input. This method improved the accuracy by up to 11% [5].

Michigan State University
Aug. 2021 - present

Research Assistant (Cont.)

COMPUTER VISION LAB

Michigan State University

Aug. 2021 - present

- Proposed a novel, training-free scoring algorithm for biometric models by combining original model scores and K-Nearest Neighbors scores which assess the presence of each query in the gallery. Conducted a comprehensive analysis of the algorithm's behavior using synthetic datasets, supported by theoretical justifications of its performance. Experimental results demonstrated up to a 10% improvement in both verification and open-set search. Prepared a research paper on this work, which is currently under submission.
- Developed a novel score normalization technique to semantically align the mean of genuine scores with the mean of imposter scores. This approach achieved a 19% total improvement across four key evaluation metrics.
- Collaborated on the development, debugging, and maintenance of a complex, end-to-end system integrating detection, restoration, and biometric modules for the 4-year IARPA-funded BRIAR project with \$12.5M in funding. Trained gait recognition models on the BRIAR dataset and successfully deployed them as containerized modules within the integrated system.

Research Intern

DR. CHENLIANG XU'S GROUP

University of Rochester

May 2020 - May, 2021

- Formulated the novel problem of separating invisible sounds from sound mixtures, which existing audio-visual sound separation methods cannot address. Designed a model that leverages video semantics to separate all sounds, with and without visible sources. Conducted extensive experiments to validate the model's effectiveness and summarized the results in a research paper [12].

Research Intern

UNDER THE SUPERVISION OF DRS. YAPENG TIAN AND CHENLIANG XU

University of Rochester

Dec 2017 - Feb 2018

- Constructed a dataset of 10-second videos of 10 musical instruments with about 60 videos for each instrument.
- Used a pre-trained inception model to extract feature representations from each video and trained a classifier using these extracted features.

Summer Research Scholar

RESEARCH SCIENCE INITIATIVE (RSI)

Beijing, China

Jul. 2016 - Aug. 2016

- Developed a convolutional neural network-based system to detect distracted drivers by analyzing visual cues of inattention.

Teaching and Mentorship

Mentorship

COMPUTER VISION LAB

Michigan State University

Sep. 2022 - present

- Detection & Tracking (MSU UG): Hank Murdock (Fall 2022), Apar Mohabansi (Fall 2022), Manh Tran (Fall 2022), Calvin DeJong (Fall 2023), Duc Hung Luong (Fall 2023)
- 3D Scene Reconstruction, Human Parsing, Person reID, etc: ZiAng Gu (MSU UG, Fall 2022 - Spring 2025)
- MSU HSHSP: Zoaib Sihorwala (Summer 2023)

Workshop Leader

TRIPODS NSF REU/GRADSTEMFORALL 2020

University of Rochester

Aug. 2020

- Co-led Python workshops to introduce foundational programming concepts, with a focus on applying Python to visualize and probabilistically verify theorems in high-dimensional probability.
- Developed and demonstrated Python scripts to simplify complex mathematical concepts through hands-on examples to enhance participants' understanding of theoretical proofs.

Teaching Assistant

VARIOUS DEPARTMENTS

University of Rochester

Sep. 2018 - Dec. 2020

- Functions of Real Variable I (Honors) (MATH 265H, Fall 2020)
- Intermediate Microeconomics (ECON 207, Fall 2020)
- Walk in CS tutoring (Spring 2020)
- Design and Analysis of Efficient Algorithms (CSC 282, Fall 2019)
- Computer Organization (CSC 252, Spring 2019)
- Computation and Formal Systems (CSC 173, Fall 2018)

Services

Server Manager

COMPUTER VISION LAB

Michigan State University

Apr. 2024 - present

- Oversee the management of 86 GPUs across 13 servers and ensure optimal performance and resource allocation for computational tasks.
- Coordinate with IT and support staff to maintain, update, and troubleshoot servers, as well as to plan and execute the procurement of new hardware.

Reviewer

FOR VARIOUS CONFERENCES

May 2022 - present

- CVPR 2025-2026, ECCV 2022-2026, ICCV 2025, NeurIPS 2024, NeurIPS 2026, WACV 2025, BMVC 2026, ICPR 2026, FG 2025-2026, IJCBLLR 2024
- CVIU (2025), TPAMI (2025)

Skills

Programming

Proficiency in Python, Java, C/C++, $\text{\LaTeX}$

Development

Familiarity with PyTorch and TensorFlow; working knowledge of Linux and command line tools

Languages

Native Chinese (Mandarin & Cantonese), fluent English, and elementary Japanese & French